Introduction to Statistical Learning

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  - Find structure without outcome labels
  - Examples: clustering, density estimation, dimension reduction

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  - Wouldn't it depend on what x are observed?

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- For sets of nonzero Boolean coordinates, Jaccard distance:
  $d_j(p, q) = \frac{|p \cup q| - |p \cap q|}{|p \cup q|}$
  Set distance to 0 when both sets are empty.

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- Great when small p, large N

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- How do we solve the problem?
- One way: parametric and structural models
  $f(x) = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + ...\beta_p * X_p$

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  - Add boundary constraint: linear before 1st knot or after last knot. (Natural spline.)

Error

- At a fixed x, let $f(x) = E[Y \mid X = x]$.
- A fresh outcome has noise variance $\sigma^2(x)$ and is independent of training data D.
- Under squared loss:

$$E_{D,Y|x}[(Y - \hat{f}_D(x))^2] = \underbrace{(E_D[\hat{f}_D(x)] - f(x))^2}_{\text{squared bias}} + \underbrace{\text{Var}_D(\hat{f}_D(x))}_{\text{variance}} + \underbrace{\sigma^2(x)}_{\text{irreducible noise}}.$$

More flexibility can lower bias and raise variance; neither pattern is universal.

Evaluating Models

- For independent observations, $L(\theta) = \prod_i p_\theta(y_i \mid x_i)$.
- Deviance: $D = 2(\ell_{\text{saturated}} - \ell_{\text{fitted}})$.
- Explained-deviance ratio: $1 - D_{\text{fitted}}/D_{\text{null}}$ when $D_{\text{null}} > 0$. This is not a universal definition of pseudo-R^2.
- For maximized log-likelihood $\hat{\ell}$ and k estimated parameters:

$$\text{AIC} = -2\hat{\ell} + 2k, \qquad \text{BIC} = -2\hat{\ell} + k \log n.$$

Compare likelihoods fitted to the same response data using consistent constants. Smaller values are preferred.

Information Criteria

- Under regularity conditions, expected test deviance exceeds training deviance by about $2k$ (same sample size).
- For Gaussian linear regression, counting variance in k:

$$\text{AICc} = \text{AIC} + \frac{2k(k+1)}{n-k-1}, \quad n > k+1.$$

- This small-sample correction is model dependent.
- BIC penalizes each parameter by $\log n$ rather than 2. It targets a different model-selection objective.

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- Balanced error rate: $1 - \frac{1}{2}(TPR + TNR)$

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- Net prediction error on test set can vary a lot.

A Clarification

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- Larger n allows for more precise estimation of relationship

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  - Valid under independence or certain positive dependence; arbitrary dependence needs other corrections.

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